A WEB-BASED PRINTING AND PHOTO SERVICES MANAGEMENT SYSTEM WITH E-MAIL NOTIFICATION FOR AMUNING TOKOBE ENTERPRISE

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APPROVAL SHEET

A System Integration and Architecture 1 project entitled **"A WEB-BASED PRINTING AND PHOTO SERVICES MANAGEMENT SYSTEM WITH E-MAIL NOTIFICATION FOR AMUNING TOKOBE ENTERPRISE"**, prepared and submitted by **SHERYL CRIS B. CARIGMA, JOHN MICHAEL C. FELIX, RONNEL O. IGNACIO,** and **JAYMEE ALYSSA R. SANTOS**, in partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY** is hereby approved and accepted.

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DEDICATION

The proponents behind this project respectfully dedicate it to God, who has been with us from the beginning and will be with us until the finish, giving us guidance, wisdom, and safety. His presence provided us the strength to face problems, the clarity to make choices, and the determination to keep going even when things were difficult. Without His blessings, none of this would have been possible.

We also offer this effort to our families, who continually supported and encouraged us. Their constant faith in our skills, their continuous reminders, and the sacrifices they made pushed us to work harder and complete this study with determination. Their love, understanding, and patience were our source of strength during times of stress and uncertainty.

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in us. To God, our families, our friends and classmates, and to Engr. Alneslyn Bucud, we dedicate our small work with thanks and appreciation.

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Chapter 1

INTRODUCTION

Background of the Study

Amuning Tokobe Enterprise is a printing and photo services business that started in August 2016 and registered under the Bureau of Internal Revenue (BIR). It is located at B22 L3 Lincoln St Corner, S. Pedracio Street, Antipolo City and began as a small business with the goal of helping the owner to pay for their expenses and bills. As time progressed, the customer demands increased and the variety of the clients expanded, the business continuously grew. Today, Amuning Tokobe Enterprise strives to be known as a one-stop printing shop that provides the request of the individuals and organizations. Printing, photo printing, the production of books and booklets, photo albums, tarpaulins and stickers are just a few of its services. They also offer photo booths for special occasions such as birthdays, graduations and weddings.

Amuning Tokobe Enterprise continues to run its operations and interactions with customers using manual methods despite its growth. G-mail, Facebook Messenger and walk-ins are the usual ways that orders and inquiries are received. Payments and expenses are manually tracked using Excel formulas, while the files of the customers are sent through G-mail. Despite being functional, this process has several disadvantages including delays in answering customer messages, problems in handling bookings and advertising for the business and the inability to handle increasing demands due to the lack of an integrated system.

The need for a more organized and easy solution becomes clear as consumer demand rises especially during special occasions such as weddings, graduations and renewal of permits. These problems can be resolved with a web-based printing and photo services management system with e-mail notifications. The enterprise can improve communication through automated e-mail updates, maintain accurate sales and customer information and improve the order processing by integrating technology into operations. In addition, having an interactive online platform would enable Amuning Tokobe Enterprise to manage bookings easier, promote its services and portfolio online and increase the number of its customers. With these improvements, the enterprise can adapt to the digital environment and maintain its image as a trusted printing and photo services business.

Problem and Objectives of the Study

General Problem

Amuning Tokobe Enterprise handles customer interactions, bookings and advertising using manual methods. This causes problems which affect client's satisfaction and limit the ability of the business to grow.

Specific Problem

- How can the enterprise reduce the delays in responding to customer inquiries and messages?
- How can the enterprise reduce the delays in responding to customer inquiries and messages?

- How can the enterprise utilize digital platforms to improve its advertisement?
- How can the enterprise organize and manage customer information to avoid errors and missed records?
- How can the enterprise provide customers with updates about their bookings and orders?

General Objective

To develop a web-based printing and photo services management system with email notification that will improve the customer communication, booking processes and services promotions for Amuning Tokobe Enterprise.

Specific Objective

- To design a system that provides faster and easy communication with customers.
- To develop a booking feature in the system that allows customers to manage their printing/photobooth request.
- By implementing digital advertisements that enhance the capabilities of the enterprise.
- To create a database that organizes and manages customer records and transactions.
- To integrate an email notification feature that provides customers with updates about their bookings and orders.

Scope and Limitations of the Study

This study focuses on the development of a web-based printing and photo services management system with e-mail notifications for Amuning Tokobe Enterprise. The system highlights the inquiry of the clients, online booking, email notifications for booking, and scheduling of their requests. Its purpose is to improve clients satisfaction and provide a better customer handling process, booking process, and entice customers through its digital promotion.

Specifically, the developed system will include:

- A public website featuring service promotions, a gallery, and contact information.
- A customer portal for submitting printing and photobooth service requests, managing their bookings, and communicating with the enterprise.
- An admin portal for enterprise staff to view, manage, and update the status of all orders and customer records.
- An automated email notification system that triggers messages for order confirmation, status updates and completion.
- A centralized database to store and manage customer information, service requests, transactions, and content for digital promotions.
- Payment gateway integration.

The study is limited to the following perimeters:

- Advanced financial accounting modules such as payroll, general ledger, or profit/loss statements.

- Inventory management for physical supplies like ink, paper, or photo frames.
- A public e-commerce shop for selling physical products unrelated to their printing and photo services.
- Features for managing multiple franchise or branch locations.

Significance of the Study

The development of web-based printing and photo services management systems with email notification is important because it addresses Amuning Tokobe Enterprise's problems especially with communication, advertising and booking. The system aims to present an interface that will improve the customer interaction while improving the business operations. This study will benefit the following:

Business Owner. By using online portfolios and displaying their services, their system can help to improve their advertising. In addition, by providing an organized booking and notification system, it will improve the response time to customer inquiries. It can lead to better customer service, more exposure and the growth of the enterprise.

Customers. The system will make it easier for the customers to check the prices, schedule services online and access the information about the services. Customers will receive current updates on their bookings and inquiries with the use of email notifications, giving them transparency and assurance that their concerns will be resolved quickly.

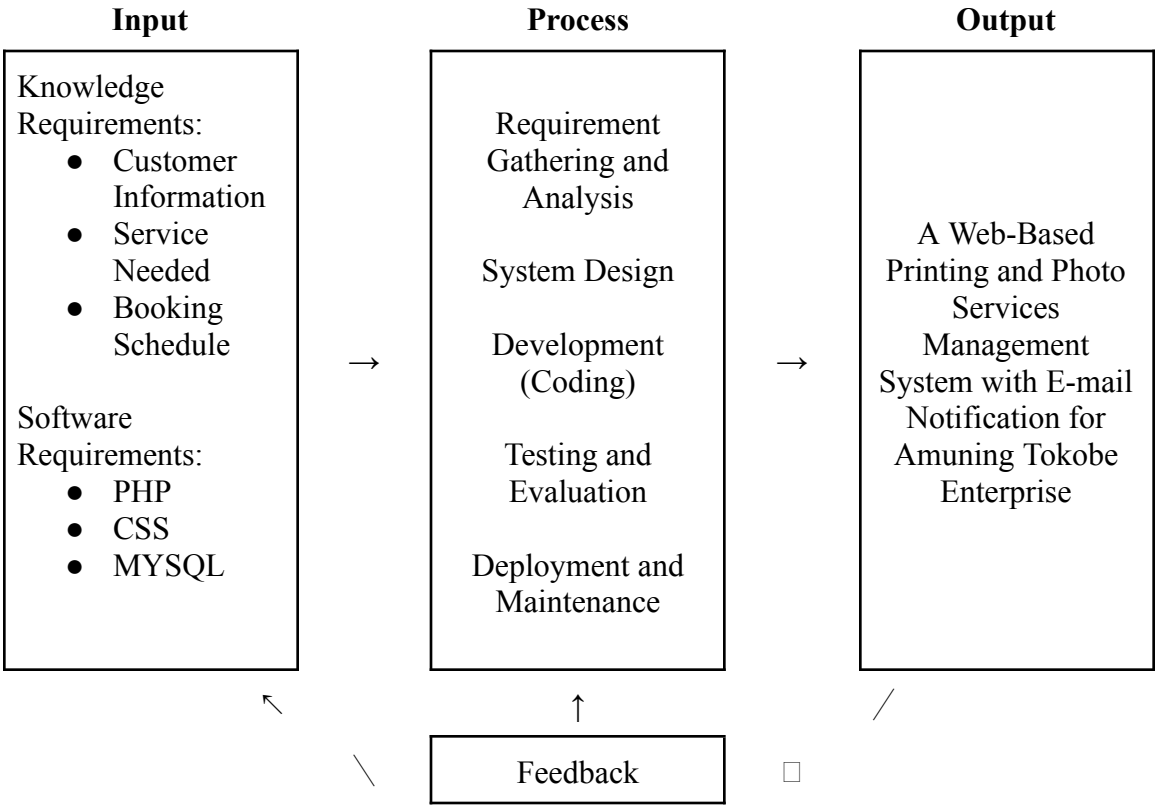
Employees. The system will reduce manual processes like communicating only via Facebook Messenger. This reduces workload, avoids messages from being ignored and allows staff to focus more on producing quality work.

Future Researchers. The study can serve as reference for researchers and future system developers who want to find solutions for small-to-medium businesses that also deal with issues in customer service, communication and advertising.

Chapter 2

CONCEPTUAL FRAMEWORK

Figure 1.0 illustrates the conceptual framework for this study, which is structured on the Input-Process-Output model. The Input component defines the essential data and system requirements needed for the development. The Process applies the Agile Software Development Life Cycle, serving as a guide to the project from initial design through to deployment and maintenance. The final Output is the delivery of a functional Printing and Photo Services Management System, which is designed with a Feedback mechanism to support ongoing system improvements and adaptations. This framework provides a clear plan for the entire project. It connects all the key stages, ensuring that the initial requirements directly inform the development process and lead to a final product that can be improved over time.



Review of Related Literature and Studies

Foreign Literature

Sustainable Digital Transformation in Small and Medium Enterprises (SMEs): A Review on Performance

Small and Medium Enterprises (SMEs) are the backbone of the global economy, 90% of all businesses and 50% of employment worldwide. Digital Transformation (DT) is defined as "a fundamental change process, enabled by the innovative use of digital technologies accompanied by the strategic leverage of key resources and capabilities, aiming to radically improve an entity and redefine its value proposition for its stakeholders" (Melo et al., 2023).

For SMEs like Amuning Tokobe Enterprise that uses Facebook Messenger, Gmail and walk-ins for handling their business are suggestive of a pre-digital transformation phase. Implementing a web-based management system represents an essential step towards digital transformation, enhancing efficiency, professionalism, and customer satisfaction. Retrieved from https://doi.org/10.1016/j.heliyon.2023.e13908

Analyzing Phase-Specific Drivers of Digital Transformation in SMEs: A Machine Learning-Based Approach

The study underscores the urgency of SMEs to adopt digital solutions to be able to remain competitive. The proposed web-based system represents a step forward towards digital transformation, which optimizes the processes of the business and enables a more

efficient method. This aligns with the study of Zhang et al. (2025) who found that in the early phase of digital transformation, SMEs primarily focus on adopting basic digital tools and technologies as a foundation for more advanced digital capabilities.

In relation to this study, Amuning Tokobe Enterprise's proposed web-based printing and photo services management system serves as the initial stage of digital transformation. By introducing an integrated digital platform for inquiries, bookings, and notifications, the enterprise began its shift from manual operations to a structured technology driven process. Retrieved from https://doi.org/10.1145/3729706.3729773

Exploring the Impact of Customer Reviews, Website Quality, Perceived Service Quality, and Product Assortment on Online Purchase Intention:The Mediating Role of Trust

Recent studies emphasize the quality and usability of a website play a crucial role in influencing a customer's trust and engagement, as a well designed platform encourages users to interact with the system more confidently. According to the findings, customers are more likely to trust and engage with a website that presents clear product information and a reliable performance, these are factors that are not often seen in basic social media platforms.

In relation to the enterprise's current reliance on fragmented platforms such as Facebook Messenger for handling inquiries and bookings, it prevents it from establishing

the professional web presence necessary to build customer loyalty and encourage repeat transactions. Developing a centralized web-based management system would therefore help the business establish a professional digital presence, enhance user confidence and promote customer loyalty. This aligns with the study's findings that modern consumers feel more comfortable in well-constructed and user-friendly systems. Retrieved from https://doi.org/10.1016/j.teler.2025.100236

Digital Payment Systems in an Emerging Economy

Ly and Ly (2024) confirmed a significant and positive relationship between the Perceived Ease of Use (PEU) of a digital payment and a user's attitude towards adopting it. According to the Technology Acceptance Model (TAM), when users perceive a system as easy to understand and operate, their attitude becomes more favorable, directly influencing their likelihood of adoption.

This suggests that system design and user perception play a crucial role in determining the success of digital platforms. In relation to this study, the proposed system is intended to be straightforward and easy for customers to use from service discovery and booking to final payment. Therefore, the integration of the payment gateway should focus not only on ensuring security but also to be user-friendly and intuitive.
Retrieved from
https://doi.org/10.1016/j.chbr.2024.100517

Digital Marketing, Financing Constraints, and Enhancements in Corporate Performance

Qiao et al. (2025) emphasized the growing importance of digital transformation in enterprise performance and competitiveness. Their study noted that in today's digital economy technological tools have become essential for organizations to optimize their processes and strengthen their connections with customers.

With regards to the proposed web-based system that aims to enhance the enterprise's visibility and competitiveness through digital promotion. The system will serve as a platform to showcase the enterprise's work and services, including printed materials and photo booth services. By featuring an online portfolio and gallery, for customers to easily browse through. Retrieved from

https://doi.org/10.1016/j.frl.2025.108221

Foreign Studies

A Comparative Study on Technological Transformation Between Automation and Manual Process

The operational inefficiencies inherent in manual processes within small businesses highlight the critical need for technological adoption. The study found that a strong majority of IT professionals perceive software automation as more efficient and faster than manual methods, primarily due to benefits like improved productivity, time-saving, and reduced human errors (Keerthana & Rajesh, 2023).

For a small enterprise like Amuning Tokoke Enterprise, which currently relies on fragmented, manual channels like social media messaging and walk-ins, these findings validate the proposed shift to an automated web-based system. The study confirms that such a system can directly address the core operational challenges faced by the business, streamlining tasks and minimizing the delays and errors characteristic of non-integrated, manual workflows. Retrieved from https://www.ijsdr.org/papers/IJSDR2308133.pdf

Printing Business Production Development Process Strategy: A Case Study of Printing UKM Nova Florist and Advertising

This need for technological improvement is a common thread across MSMEs, whose resilience and strategic development are critical to national economic stability, as they contribute significantly to growth and employment, even during periods of economic crisis (Iba et al., 2022). It was studied that effective strategy formulation is essential to navigate competition and ensure sustainability.

The study suggests that business development can be pursued through key strategies such as capital management, labor optimization, and maintaining strong customer service and relationships. Furthermore, the production and operations function, which encompasses all activities transforming inputs into finished goods, must be meticulously managed as it often represents the largest portion of a company's assets. For Amuning Tokobe Enterprise, applying these strategic principles to its diverse product line, from floral boards to digital printing, is fundamental to overcoming operational

challenges and achieving long-term viability in the local market. Retrieved from https://doi.org/10.54443/ijset.v2i1.134

The Challenges and Strategies of Printing Industry

The strategic imperatives for MSMEs are magnified by the specific pressures facing the global printing industry, which is at a critical juncture, compelled to adapt to profound shifts in customer behavior and rapid technological advancements to ensure its survival and growth. The Challenges and Strategies of Printing Industry by Amri et al. (2022) identifies a convergent set of core challenges across Indonesia and Malaysia, primarily stemming from the digitalization era which reshapes market demand and necessitates technological upgrades. These challenges are consistently categorized into four key areas: fluctuating market demand, the pace of technological development, the management of human resources, and the imperative of maintaining high product quality.

In direct response to these challenges, strategic adaptations are essential. For a specific enterprise like Amuning Tokobe Enterprise, this means that strengthening marketing efforts, investing in new printing technologies like digital and offset presses, implementing continuous skill-based training for employees, and enforcing rigorous quality management systems are no longer optional. These strategies are vital to build customer trust and maintain a competitive edge in a rapidly evolving market.

Retrieved from https://doi.org/10.4108/eai.16-11-2022.2326107

An Exploratory Study on Customer Relationship Management Implementation and Strategies

The successful implementation of a Customer Relationship Management (CRM) strategy is paramount for converting customer engagement into tangible sales performance, yet a gap often exists between strategic goals and actual outcomes. An Exploratory Study on Customer Relationship Management Implementation and Strategies by Sardjono & Tanuwijaya (2024) analyzed a retail company facing a sales target shortfall despite using CRM, identifying five critical factors influencing this performance gap: CRM Reliability, Underutilization of CRM, CRM Capability, CRM Unpreparedness, and Customer Relation's Quality. Their developed regression model demonstrated that factors like Reliability, Capability, and Relation's Quality positively impact performance, while Underutilization and Unpreparedness have a significant negative effect.

For a business like Amuning Tokobe Enterprise that is implementing a customer-centric strategy, these findings underscore that simply having a CRM system is insufficient; success hinges on ensuring the technology is reliable, that staff are fully trained and capable, that the system is fully utilized rather than underused, and that the organization is thoroughly prepared to leverage customer data to build high-quality, personal relationships that drive repeat purchases and bookings. Retrieved from https://doi.org/10.33168/JSMS.2024.0310

OECD Studies on SMEs and Entrepreneurship The Digital Transformation of SMEs

Finally, the digital transformation presents a critical pathway for Small and Medium-sized Enterprises (SMs) to achieve these goals of operational efficiency and customer satisfaction, yet a significant adoption gap persists compared to larger firms (OECD, 2021). As noted in their study, SMEs tend to lag in digital uptake, often due to barriers such as limited internal resources, skills gaps, and financing constraints. This lag is particularly pronounced with more sophisticated technologies, while adoption is higher for basic functions like administration and marketing.

For a business like Amuning Tokobe Enterprise, which relies on fragmented manual processes, this underscores the strategic importance of a targeted digital solution. The OECD further highlights that the COVID-19 crisis acted as a catalyst, accelerating SME digitalization and demonstrating that integrated, web-based systems are no longer a luxury but a necessity for operational resilience, improved customer communication, and sustainable growth in the modern economy. Retrieved from https://www.oecd.org/content/dam/oecd/en/publications/reports/2021/02/the-digital-transformation-of-smes_ec3163f5/bdb9256a-en.pdf

Local Literature

Web-Based Ordering System for Start-Up Business with Forecasting

A Web-Based Ordering System for Start-Up Businesses with Forecasting was developed by Labastida, Simbulan, Estrella, Flores, and Centeno (2024) to improve operational effectiveness and facilitate data-driven decision-making. To prepare for client demand and maximize resource allocation, their system integrated forecasting tools. This

study demonstrates how small businesses may improve inventory control and service consistency through the use of automation and analytical skills.

Such forecasting capabilities can be adapted for Amuning Tokobe Enterprise, which offers both printing and photo services, to estimate peak service demands such as around graduation or event seasons ensuring that resources, supplies, and employees are handled properly. The study shows how important it is to integrate predictive analytics into web-based business systems in order to maintain precision, effectiveness, and customer satisfaction. Retrieved from https://doi.org/10.30574/wjarr.2024.22.3.1721

The Challenges of Small Businesses in Today's Economy

Small businesses are widely recognized as the backbone of the Philippine economy, yet they face a multitude of challenges that threaten their sustainability. According to an analysis by All Properties (2022), these obstacles include intense competition from larger corporations with greater brand recognition and marketing budgets, difficulties in accessing capital, and the rising pressure to adapt to a digital economy dominated by e-commerce and online competitors. To overcome these hurdles, the literature suggests that small businesses must leverage their unique advantages, such as providing superior customer service, forming strategic partnerships, and utilizing technology to automate tasks and improve operational efficiency.

In a similar context, Amuning Tokobe Enterprise faces these very challenges in a

competitive printing and photo services market. By developing a web-based management system, the enterprise directly addresses these issues by creating a digital storefront to compete more effectively, automating manual tasks like order tracking and confirmation to improve efficiency, and providing a centralized platform for customer interaction to enhance service quality. This strategic use of technology, as recommended in the literature, can help the enterprise overcome common small business constraints, solidify its market position, and build a more resilient and modern operation. Retrieved from https://www.allproperties.com.ph/the-challenges-of-small-businesses-in-todays-economy/

Development of a Web-Based Food Ordering System

Jiangmin (2024) addressed in "Development of a Web-Based Food Ordering System" how digital ordering systems improve customer and business communication and improve backend operations. The system featured variables like order tracking, receipts and consumer feedback tools despite its primary focus on the food business.

These features are directly related to Amuning Tokobe Enterprise where customers who submit orders for printing or photography can receive digital updates and confirmations via email notifications. This ensures communication in real time, minimizes miscommunication and offers an organized transaction record. The study shows that customer satisfaction and efficiency are greatly increased by transparent, automated digital systems which are the goals of Amuning Tokobe's proposed management system. Retrieved from https://doi.org/10.70979/FRNW7062

The Impact of E-Commerce on Small and Medium Enterprises in the Second District of Albay

The adoption of e-commerce is a critical factor for the growth and competitiveness of Small and Medium Enterprises (SMEs), as demonstrated by Flores (2025) in a study on SMEs in the Second District of Albay. The research found that the integration of e-commerce had a positive impact on key business metrics such as revenue growth, market reach, and cost efficiency. However, the study also identified significant barriers to adoption, including a lack of technical skills, inadequate financial resources, and concerns over privacy and security. These findings highlight that while the potential benefits of digital transition are clear, successful implementation requires overcoming specific, localized challenges faced by SMEs.

Similar to the Albay SMEs, Amuning Tokobe Enterprise can leverage these insights to guide its own digital transition. By recognizing that technical capability and customer trust are paramount, the enterprise can prioritize the development of a user-friendly, secure, and reliable web-based system for managing printing and photo services. Ensuring the platform is easy to use addresses the skill barrier, while robust data protection measures can alleviate privacy concerns, thereby encouraging adoption and helping the enterprise to achieve the same benefits of expanded market reach and improved operational efficiency identified in the literature. Retrieved from https://rsisinternational.org/journals/ijriss/articles/the-impact-of-e-commerce-on-small-and-medium-enterprises-in-the-second-district-of-albay/

Philippines Printers & Copiers Market (2025-2031): Dominating Trends & Future Growth

The printing market in the Philippines is set for steady growth, driven by new technology and business needs. According to Philippine Duplicators Inc. (2025), the market is expected to grow steadily, reaching USD 74.7 million by 2025. Key trends include the rise of digital and on-demand printing, a focus on eco-friendly practices, and smarter multifunction printers with AI. For businesses, this creates opportunities to serve growing SMEs, offer new types of printing services, and provide strong customer support to succeed in the evolving market.

Similar to this, Amuning Tokobe Enterprise can leverage these market insights to guide the development of its services. By aligning its offerings with the identified trends, such as adopting on-demand printing capabilities, promoting sustainable printing options, and integrating smart, automated features into its service workflow, the enterprise can position itself as a modern and competitive player. Focusing on the growing SME segment and prioritizing robust customer support, as highlighted in the literature, will allow Amuning Tokobe Enterprise to effectively capture market opportunities and ensure long-term growth. Retrieved from https://phildup.com/future-printers-philippines-2025-2031/

Local Studies

Determinants of Demand For Photography Services

In the Philippine setting, the persistence of manual and paper-based systems among small and medium enterprises (SMEs) highlights the ongoing need for digital transformation. Enema and Quezon (2024) examined the determinants of demand for photography services, emphasizing how digital solutions can enhance accessibility and efficiency in service delivery.

For small enterprises like Amuning Tokoke Enterprise, which currently depend on manual operations such as handwritten logs or social media communication, these findings affirm the importance of adopting integrated web-based systems. The implementation of automation not only enhances productivity but also ensures accuracy and faster turnaround in business transactions key factors in maintaining competitiveness in a growing digital economy. Retrieved from https://www.researchgate.net/publication/384451980_Determinants_of_Demand_for_Photography_Services

Production Management of Print Advertising Services in Nueva Ecija

Likewise, Armas and Villegas (2022) examined the Production Management of Print Advertising Services in Nueva Ecija, Philippines, emphasizing operational discipline in scheduling, material management, and quality control. Their study highlights how the local printing industry continues to face production inefficiencies due to lack of centralized systems.

The study suggests that efficient production management in print advertising enhances workflow and client satisfaction. It finds that by applying automated management tools, printing businesses can streamline operations, reduce idle time, and optimize resource allocation to ensure both timely service completion and consistent quality. These findings collectively reinforce that implementing such systematic, digitized methods in a web-based platform allows small printing enterprises to optimize their entire production cycle and significantly improve customer satisfaction.

Retrieved from

https://doi.org/10.21275/SR15122151102

Level of Awareness of Smart Manufacturing Technologies and Its Nexus to Adoption Among Micro, Small, and Medium Enterprises in the Philippines

Further expanding on the technological readiness of Filipino MSMEs, Barroga, Wabina, and Sales (2021) analyzed the Level of Awareness of Smart Manufacturing Technologies and its adoption among local enterprises. Their study revealed that while awareness of digital tools such as automation, AI, and real-time monitoring is increasing, actual implementation remains limited by cost and expertise barriers. This situation underscores the relevance of developing simpler, cost-efficient web-based systems that address the unique operational needs of MSMEs.

The study implies that awareness and adoption of digital technologies among MSMEs are vital to improving operational efficiency, suggesting that incorporating smart

and automated processes into a web-based system promotes productivity and modernizes traditional business practices. For a business like Amuning Tokoke Enterprise, this validates the necessity of accessible automation, demonstrating how modernization can be supported without demanding extensive financial or technical resources.

Retrieved from https://philjournalsci.dost.gov.ph/wp-content/uploads/2025/07/level_of_awareness_of_smart_manufacturing_technologies_and_its_nexus_.pdf

Interactive Printing Kiosk for Aemilianum College Inc

Innovation at the institutional level further supports this transition. Aban and Sarmiento (2025) introduced an Interactive Printing Kiosk for Aemilianum College Inc., enabling users to independently process print jobs through an automated interface. This system significantly reduced transaction time and enhanced service accessibility, reflecting how local digital solutions can efficiently bridge gaps in user service and operational management.

This is relevant to the study's implication that integrating automated printing systems, like an interactive kiosk, can streamline transactions and reduce manual workload for a business like Amuning Tokobe Enterprise. This approach ensures accuracy, faster service, and improves customer experience, which are key goals of a web-based printing management system. For enterprises like Amuning Tokobe exploring automation, such systems serve as practical and transformative models for streamlining daily business functions. Retrieved from

https://www.globalscientificjournal.com/researchpaper/INTERACTIVE_PRINTING_KIOSK_FOR_AEMILIANUM_COLLEGE_INC_.pdf

Effects of Photo Editing Towards Electronic Commerce in Metro Manila

Finally, Pablo et al. (2022) explored the Effects of Photo Editing towards Electronic Commerce in Metro Manila, demonstrating how digital presentation directly influences consumer engagement and purchasing decisions. For printing and creative businesses, this finding underscores the strategic value of integrating digital tools to enhance product appeal and communication with clients.

The study implies that the use of photo editing and digital tools significantly improves online business engagement. For Amuning Tokobe Enterprise, incorporating these visually engaging and efficient digital processes into its new web-based system is essential. This integration can boost marketing effectiveness, attract more customers through visually appealing outputs, and improve the overall user experience, thereby helping the business maintain its competitiveness in a tech-driven marketplace.

Retrieved from

https://al-kindipublisher.com/index.php/jbms/article/view/3175

Chapter 3

RESEARCH METHODOLOGY

This chapter explains the research approach used to create A Web-Based Printing and Photo Services Management System with E-mail Notification for Amuning Tokobe Enterprise. It describes the processes and procedures utilized to collect, analyze, and interpret data required for the design and implementation of the proposed system.

Methodology of the Study

The plan, strategy, and processes required to carry out the inquiry and find a reliable solution to the recognized problem are provided by the methodology, which is an essential part of any research study. Amuning Tokobe Enterprise's Web-Based Printing and Photo Services Management System with E-mail Notification was developed using the Agile Development Methodology. This adaptable, iterative approach allows for continual improvement throughout the system's life.

Agile was selected because it enables researchers to include user feedback throughout the development process while designing, creating, testing, and improving the system in brief, manageable cycles. Through this technique, the researchers were able to quickly develop system features, solve problems as they emerged, and meet expectations. Each phase of the Agile approach helped us create a system that is functional, user-friendly, and meets Amuning Tokobe Enterprise's operational objectives.

The phases of the Agile process used in this investigation are requirements, design, development, testing, deployment, and review. Every stage helped turn the collected data into a fully functional web-based printing and photo services management system with integrated email notification capabilities.

Phase 1 – Requirements. This phase is to identify and gather all system requirements. Researchers analyzed the needs of the business owner, staff, and customers using surveys, interviews, and document analysis. Manual processes such as order handling, customer communication, and record-keeping were examined to define the system's functional and non-functional requirements.

Phase 2 – Design. This phase includes the development of the system's overall design and organization. This entailed creating the system architecture, database structure, and

user interface. Wireframes and prototypes were developed for the public website, customer portal, and admin portal to ensure intuitive navigation and user experience.

Phase 3 – Development. During this phase, the system was actually coded and implemented. This phase included the development and implementation of critical functionality such as online booking, service inquiry management, customer database, automated email notifications, and payment gateway integration using PHP, MySQL, HTML, CSS, and JavaScript.

Phase 4 – Testing. System testing identified and fixed vulnerabilities, performance concerns, and problems. Several tests, including security, usability, and functionality tests, were conducted to make sure that every part of the system operated as intended. User acceptance testing was performed with the client to validate system performance against business requirements.

Phase 5 – Deployment. The system was put into operation at Amuning Tokobe Enterprise. Installation, configuration, and initial user training were carried out to ensure optimal system operation. The system was hosted on a live web server and made accessible to customers and staff.

Phase 6 – Review. The last phase includes system review and continual improvement. The efficacy, usability, and performance of the system were evaluated by collecting user feedback. The system was updated and upgraded as necessary to guarantee its effectiveness, security, and flexibility in response to changing needs based on the evaluation's conclusions.

REFERENCES

Journals and Publications

Aban, J. S., & Sarmiento, J. R. (2025). *INTERACTIVE PRINTING KIOSK FOR AEMILIANUM COLLEGE INC*. Global Scientific Journal, 13(2). https://www.globalscientificjournal.com/researchpaper/INTERACTIVE_PRINTING_KIOSK_FOR_AEMILIANUM_COLLEGE_INC_.pdf

Amri, L., Farani, D., Ali, N., & Anwar, R. (2022). *The Challenges and Strategies of Printing Industry*. Proceedings of the First Jakarta International Conference on Multidisciplinary Studies towards Creative Industries, JICOMS 2022, 16 November 2022, Jakarta, Indonesia. https://doi.org/10.4108/eai.16-11-2022.2326107

Armas, K. L., & Villegas, M. N. (2022). *PRODUCTION MANAGEMENT OF PRINT ADVERTISING SERVICES IN NUEVA ECIJA, PHILIPPINES*. Academia.edu. https://www.academia.edu/105071783/Production_Management_of_Print_Advertising_Services_in_Nueva_ECIJA_Philippines

Barroga, K. D., Wabina, V. E. R., & Sales, A. C. (2021). *LEVEL OF AWARENESS OF SMART MANUFACTURING TECHNOLOGIES AND ITS NEXUS TO ADOPTION AMONG MICRO, SMALL, AND MEDIUM ENTERPRISES IN THE PHILIPPINES*. Philippine Journal of Science, 150(4), 1171–1183. https://philjournalsci.dost.gov.ph/wp-content/uploads/2025/07/level_of_awareness_of_smart_manufacturing_technologies_and_its_nexus_.pdf

Enema, R., & Quezon, M. (2024). *DETERMINANTS OF DEMAND FOR PHOTOGRAPHY SERVICES*. Journal of Interdisciplinary Perspectives, 2(11), 132–140. https://www.researchgate.net/publication/384451980_Determinants_of_Demand_for_Photography_Services

Flores, J. (2025). *The Impact Of E-Commerce On Small And Medium Enterprises In The Second District Of Albay - International Journal of Research and Innovation in Social Science.* https://rsisinternational.org/journals/ijriss/articles/the-impact-of-e-commerce-on-small-and-medium-enterprises-in-the-second-district-of-albay/

Hanaysha, J. R., Ramadan, H. I., & Alhyasat, K. M. K. (2025). *Exploring the Impact of Customer Reviews, Website Quality, Perceived Service Quality, and Product Assortment on Online Purchase Intention: Mediating Role of Trust*. Telematics and Informatics Reports, 100236. https://doi.org/10.1016/j.teler.2025.100236

Iba, Z., Abdur Rasyid Zasri, Durrah Safira, Ifvac Vara Azzerah, & Riska Munanda. (2022). *PRINTING BUSINESS PRODUCTION DEVELOPMENT PROCESS STRATEGY (CASE STUDY OF PRINTING UKM NOVA FLORIST AND ADVERTISING, LHOKSEUMAWE CITY)*. International Journal of Social Science Educational Economics Agriculture Research and Technology (IJSET), 2(1), 1077–1090. https://doi.org/10.54443/ijset.v2i1.134

Jiangmin, L. (2024). *Development of a Web-based Food Ordering System*. Asia Pacific Journal of Management and Sustainable Development, 12(2 Part 3), 152-156. https://doi.org/10.70979/FRNW7062

Kumar, R. & Keerthana, R. (2023). *A Comparative study on Technological transformation between Automation and Manual process*. International Journal of Scientific Development and Research, 8, 909. https://www.ijsdr.org/papers/IJSDR2308133.pdf

Online and Multimedia Resources

Aeizon. (2024). *The Benefits of Web-Based Systems for Business*. https://www.aezion.com/blogs/the-benefits-of-web-based-systems-for-business/

All Properties. (2022, April 13). *The Challenges of Small Businesses in Today's Economy*. All Properties. https://www.allproperties.com.ph/the-challenges-of-small-businesses-in-todays-economy/

Android Developers. (n.d.). *About notifications*. Retrieved from https://developer.android.com/develop/ui/views/notification

AT-Automation. (n.d.). *Webbased*. Retrieved from https://www.at-automation.nl/en/knowledge-base/webbased/

cobrief. (2025, April 2). Printing: Overview, definition, and example. Retrieved from https://www.cobrief.app/resources/legal-glossary/printing-overview-definition-and-example/

DHL Freight Connections. (n.d.). *Booking*. Retrieved from https://dhl-freight-connections.com/en/logistics-dictionary/booking/

Dwivedi, S. (n.d.). What is Service? Its Features, Types, Importance & Examples. Retrieved from https://plutuseducation.com/blog/what-is-service/

Entre. (2024, July 30). *The Impact of Inefficient Technology on Business Growth*. Retrieved from https://www.entremt.com/the-impact-of-inefficient-technology-on-business-growth/

ESA. (2009, January 28). *What is technology?*. Retrieved from https://www.esa.int/Enabling_Support/Space_Engineering_Technology/What_is_technology/

Gordon, B. (2025, May 28). *15 benefits of a website for your business*. Retrieved from https://www.wix.com/blog/benefits-of-a-website

Hayes, A. (2025, May 15). *What Is a Business? Understanding Different Types and Company Sizes*. Retrieved from https://www.investopedia.com/terms/b/business.asp

inLight. (n.d.) What is a Photo Booth? (and why you need one at your next event). Retrieved from https://www.inlightphotobooths.com/what-is-a-photo-booth/

Jolene. (2023, September 28). *What is the difference between "picture" and "photo"?*. Retrieved from https://medium.com/@Lancee/what-is-the-difference-between-picture-and-photo-b0c7605bb0ce

Kenton, W. (2025, April 9). *Customer: Definition and How to Study Their Behavior for Marketing*. Retrieved from https://www.investopedia.com/terms/c/customer.asp

APPENDIXES

Appendix A: Transcript of Interview

Interviewer: What is the name of your business?

Interviewee: Amuning Tokobe Enterprise

Interviewer: What are your main products or services?

Interviewee: Photobooths, Printing services (printing, photo printing, booklets, books, photo album, tarpaulin, stickers, etc.)

Interviewer: How long has your business been operating?

Interviewee: Since August 2016, registered BIR

Interviewer: How did your business start, and what inspired you to start this kind of business?

Interviewee: PLDT, nagwwork sa bank need pambayad ng internet and may idea sa printing so he started a business

Interviewee: Nagstart sa area nila then lumago ang market.

Interviewee: Gusto lang mabayaran bills sa pldt kaya nagstart. (Finances)

Interviewer: What is the main purpose and goal of your business?

Interviewee: One-stop printing shop, anything na hingin and tanungin meron sila.

Interviewee: lumawak pa yung variety of clients.

Interviewer: Where is your business located?

Interviewee: B22 L3 Lincoln st cor, S. Pedracio Street, Antipolo City, 1870 Rizal

Interviewee: Cuesta verde village

Interviewer: What are your business hours?

Interviewee: 8 am up to 8pm

Interviewer: Do you have any other branches?

Interviewee: Wala na, meron before (pandemic) pero clinose.

Interviewee: Kesa mag open ng bagong branch nag focus na lang sila online.

Interviewer: Who are your typical customers (individuals, businesses, events)?

Interviewee: Schools, individuals near the subdivision,

Interviewer: How do you connect or engage with your target market?

Interviewee: Mainly messenger since nakakapag send sila ng template

Interviewer: What devices or equipment do you use in running your business?

Interviewee: Office, Photoshop, Viber, Gmail, Printers, and Camera.

Interviewer: We've noticed that your business offers photobooths, do you exclusively do walk-ins, or do you also offer photobooths for occasions (e.g. birthdays)?

Interviewee: Offers photoboots for occasions.

Interviewer: Do customers ever ask about your services online or request an online portfolio or price list?

Interviewee: Yes, para malaman nila kung ano ung kaya ng business i-offer.

Interviewer: Do you require deposits or upfront payments for bookings? If yes, how do you usually collect them?

Interviewee: Yes, meron silang list then once na nag approve na sa pricing, dun sila maggo.

Interviewer: Do you notice certain months or seasons when there's an increase in customer

demand?

Interviewee: Sa printing, january to february maraming nag aasikaso ng permit since online na pag rerenew

Interviewee: March to april graduation so diplomas and lanyards ng medal

Interviewee: Bermonths more on weddings .

Interviewer: Have you faced any challenges so far? If yes, how did you overcome them?

Interviewee: Puhunan isa sa problem, when u have a business malaki project tas may kulang, and may specific na makina na hindi ma-avail, nag loan na lang sa bank ung may specialized loan for small business. The other problem is the communication with customers.

Interviewer: How do you currently store and manage your business data?

Interviewee: Through gmail

Interviewer: How do you monitor your daily sales?

Interviewee: By using excel's formula. Naicocompute magkano pumasok, magkano gastos.

Interviewer: What payment methods do you accept?

Interviewee: Bank transfer, Gcash , Cash

Interviewer: Do you have any social media accounts to promote your business?

Interviewee: Fb lang, di nila masyado nappromote kasi dalawa lang sila nag hahandle and lumalaki na rin business talaga nila.

Interviewer: Do you have an existing system/website for your business? If none, have you

considered creating a website in the past? Why or why not?

Interviewee: No, since when we were starting out, wala pang vision as to how it will

benefit them since small palang business nila, but they are open for

changes now.

Interviewer: Will you allow us to develop a system for your business?

Interviewee: Yes

Interviewer: If you were to have a website, what features would you like to see that are also useful to you?

Interviewee: As long as ma-ipakita lang ung capabilities nung business nila.

Interviewer: Would you like to showcase your portfolio or past work on the website?

Interviewee: Yes

Interviewer: Would you prefer a simple informational site, or something more interactive (e.g.,

bookings, orders, chat support)?

Interviewee: Interactive system since isa sa problem namin, like what was said before minsan yung communication with the customers, meron kaming hindi

nababasa na messages.

Interviewer: Are there any branding materials (like logos, colors, photos) you already have that we could use?

Interviewee: Yes, you are allowed to use the names, logos, and pictures.

Interviewer: How involved would you like to be in the design or development process?

Interviewee: We are open for everything, kung anong kailangang information para sa project.

Interviewee: You can create group chats para mas madali makipag communicate.

Appendix B. Comments and Suggestions

Preliminary

- Frequently Asked Questions and Built-in Messenger they are combined to form an interactive chat interface which customers can type their questions in the chat box and received instant automated responses
- Calendar Booking System allows customers to select their preferred booking dates and times for services
- Google Sign-in Option, a login feature to make user registration and sign-in faster and more convenient
- Payment Methods such as Gcash
- Terms and Conditions Page which protect the business legally and ensure transparency between the company and its customers

Midterms

- Provide a printable receipt sent to the customer's email a few hours after they make a booking
- Allow customers to book other available time slots since the owner may have multiple open hours (availability is based on hourly, not daily)
- Implement in-app notifications for new orders
- Add a semi-chatbot for FAQ
- A thread section so the admin can communicate directly with the client
- Use a colored version of the logo
- Remove redundant alerts (e.g., repeated sign-in messages)

- Integrate an online payment method

- Add login attempt limits

- Add verification code attempt limits

- Reduce the wait time for receiving the verification code via Gmail

- Implement stronger password security measures (e.g., combinations of characters)

- Show Terms and Conditions before creating an account, including refund policy

- Once the user is logged in, the name fill-up fields should no longer appear to avoid redundancy

- Fix the layout for printing service and photo service sub-menu

- User must be able to manage their orders, and update their information

RESEARCHER'S PROFILE